

# Assignment - 1.1

**Q - 1.** Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



**Q - 2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases. Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).**

| <b>Feature</b>               | <b>HDD<br/>(Hard Disk Drive)</b>                                                 | <b>SSD<br/>(Solid State Drive)</b>                              |
|------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Speed</b>                 | Slower<br>( 80 – 150 MB/s )                                                      | Much Faster<br>( 500 – 7000+ MB/s )                             |
| <b>Durability</b>            | Less Durable<br>( Contains moving parts,<br>susceptible to shock and<br>damage ) | More Durable<br>( No moving parts,<br>shock-resistant )         |
| <b>Noise</b>                 | Can be noisy                                                                     | Silent                                                          |
| <b>Power<br/>Consumption</b> | Higher                                                                           | Lower                                                           |
| <b>Capacity</b>              | Higher capacities available<br>( 1TB, 2TB, 4TB , 8TB+ )                          | Lower cost per GB but<br>improving ( 512GB, 1TB,<br>2TB, 4TB+ ) |
| <b>Typical<br/>Use Cases</b> | Mass storage, backups,<br>large files, older PCs                                 | OS, gaming, editing,<br>laptops, high<br>performance PCs        |

**Q - 3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.**

**CPU (Central Processing Unit) :-**

The CPU acts as the brain of the system, handling the game's core logic, physics calculations, and artificial intelligence. In battle royale games like PUBG or Free Fire, it constantly tracks player positions, bullet trajectories, and network data while sending instructions to the rest of the hardware.

**RAM (Random Access Memory) :-**

RAM serves as the short-term memory that holds active game data so the CPU can access it instantly. It temporarily stores the map layout, player skins, and audio files currently in use, preventing the game from stuttering or constantly lagging while loading assets from your storage drive.

**GPU (Graphics Processing Unit) :-**

The GPU is dedicated to rendering everything you see on the screen, translating data from the CPU into 3D images. It calculates lighting, shadows, textures, and frame rates, ensuring that the fast-paced action and environments of the game look sharp and run smoothly.

**Q - 4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.**

| <b>Hardware Component</b>          | <b>Why Upgrade It?</b>                                                                                                                                  |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CPU<br/>(Processor)</b>         | A faster processor improves overall game performance, reduces lag, and helps the game handle AI, physics, and background tasks efficiently.             |
| <b>GPU<br/>(Graphics Card)</b>     | The graphics card has the biggest impact on gaming performance. Upgrading it allows higher frame rates, better graphics quality, and smoother gameplay. |
| <b>RAM<br/>(Memory)</b>            | Increasing RAM (at least 16 GB recommended) helps the game load faster, reduces stuttering, and allows multitasking without slowing down the system.    |
| <b>SSD<br/>(Solid State Drive)</b> | Installing the game on an SSD greatly reduces loading times and improves overall system responsiveness compared to an HDD.                              |
| <b>Power Supply<br/>(PSU)</b>      | If you install a more powerful graphics card or processor, you may need a higher-wattage PSU to provide stable power.                                   |

|                                   |                                                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cooling System</b>             | Better cooling (extra fans or a better CPU cooler) prevents overheating during long gaming sessions and helps maintain performance. |
| <b>Motherboard (if necessary)</b> | If the new CPU or RAM is not compatible with the current motherboard, upgrading the motherboard may also be required.               |